

System Requirement Specification - Quasi-Static Magnetic Circuit

Released Rev A. Parameter values in later approved change records override conflicting values in this document.

1. System purpose

The benchmark magnetic circuit consists of a rectangular iron core with a square cross section, one discrete air gap, an exciting coil, a measuring coil located at the air-gap region, a useful-flux sensor, and a leakage-flux path. It is modeled using quasi-static magnetic flux tubes under a linear-reluctance assumption.

2. Functional behavior

A positive RMS current-phasor magnitude ramp through the exciting coil produces a positive magnetomotive force $N \cdot I$. The resulting total core flux follows the defined positive magnetic direction. At the right-leg branch, total core flux divides into useful air-gap flux and leakage flux and rejoins before the lower yoke.

Flux density B is derived from flux Φ and cross-sectional area. Magnetic field strength H is derived from B using the applicable permeability. Magnetic potential difference is H times mean flux-line length. The series magnetic potential drops plus the parallel air-gap/leakage branch drop balance the exciting-coil mmf.

3. Released nominal geometry

Item	Value	Notes
a	25 mm	Square iron cross-section side
l	150 mm	Reference mean geometry dimension from drawing
delta	1.50 mm	Nominal air-gap length
Left/upper/lower iron length	$l - a$	Mean dashed flux line
Right-leg iron length	$l - a - \text{delta}$	Air gap removed from iron length

4. Quasi-static and induction requirements

Flux, flux density, field strength, magnetic potential difference and electric quantities are represented by complex phasors. For the released linear elements, reluctance is real, so flux and magnetic potential difference share phasor angle.

The exciting coil induced voltage has positive defined electric orientation; the measuring-coil voltage has negative orientation relative to the useful air-gap flux. Both follow the quasi-static relation $|U| = 2 \cdot \pi \cdot f \cdot N \cdot |\Phi|$.

5. Grounds and assumptions

An electric ground and a magnetic ground shall both be present to establish reference potentials. Force interaction is out of scope. The core is linear in this benchmark; nonlinear saturation is handled only by a separate model variant and shall not be inferred into the baseline.